Assignment 4 - Absolute and Relative Paths in Linux

The objective of this lab is to familiarize yourself with absolute and relative paths in Linux, and to gain practical experience in navigating the file system using command-line tools.

Ensure that you have access to a Linux terminal for the following exercises.

Questions for this assignment

1. Please provide a practical explanation of absolute paths, including an example.
2. Please provide a practical explanation of relative paths, including an example.
3. How do you do Path Manipulation?

Solution by the Instructor:

1. Please provide a practical explanation of absolute paths, including an example.
    a. Open the terminal.
    b. Use the pwd command to display the current working directory.
    c. Create a directory structure with multiple nested directories using the mkdir command. For example:
    mkdir -p ~/labs/project_lab/dir1/dir2/dir3
    d. Use the cd command to navigate to the deepest directory (dir3).
    e. Use the pwd command again to verify your current location.
    f. Write down the absolute path of the current directory.

2. Please provide a practical explanation of relative paths, including an example.
    a. Navigate back to the starting directory using a relative path. Hint: Use cd ...
    b. Create a file inside dir1 using the touch command. For example:
    touch ~/labs/project_lab/dir1/file.txt
    c. Navigate to dir2 using a relative path.
    d. Use the ls command to list the contents of dir2.
    e. Write down the relative path to the file you created in dir1 from the current directory.

3. How do you do Path Manipulation?
    a. Navigate back to the home directory using the cd command.
    b. Use the ls command to list the contents of your home directory.

c. Create a symbolic link to the file you created in dir1 using the ln command. For example:
   ln -s ~/labs/project_lab/dir1/file.txt ~/link_to_file
d. List the contents of the home directory again and observe the symbolic link.
e. Use the cat command to display the content of the file through the symbolic link.